In medical applications, BVLOS drones equipped with 5G communication modules have been used to transport vaccines, blood products, diagnostic samples, critical medicines, and life-saving emergency supplies. The high-bandwidth and low-latency characteristics of 5G networks ensure continuous telemetry monitoring, real-time route correction, and stable command-and-control links throughout the mission. These capabilities reduce response time and expand access to emergency healthcare services, especially in mountainous and tribal regions.

In agriculture, BVLOS drones have been deployed for crop monitoring, soil moisture evaluation, fertilizer and pesticide spraying, and crop health analytics. Using real-time aerial imagery streamed over 5G networks, farmers and agri-scientists can make data-driven agronomic decisions, enabling more precise resource usage and improved crop productivity. The extended range of BVLOS operations allows coverage of large farmland regions with fewer takeoffs and landings, making drone-based agriculture more economically viable.

These trials also contribute to the establishment of a national unmanned traffic management (UTM) framework under the Digital Sky platform, supporting automated flight approvals, geofencing, and safe airspace integration. The results highlight India’s strategy of using telecom-enabled UAV networks to address challenges in healthcare accessibility and agricultural efficiency.

While the progress is promising, continued development is needed in battery endurance, autonomous obstacle avoidance, rural connectivity reliability, and operator training. Nonetheless, India’s 5G BVLOS trials represent a significant step toward scalable drone-based public service networks, positioning the country as an emerging leader in real-world drone utility deployment across social and economic sectors.